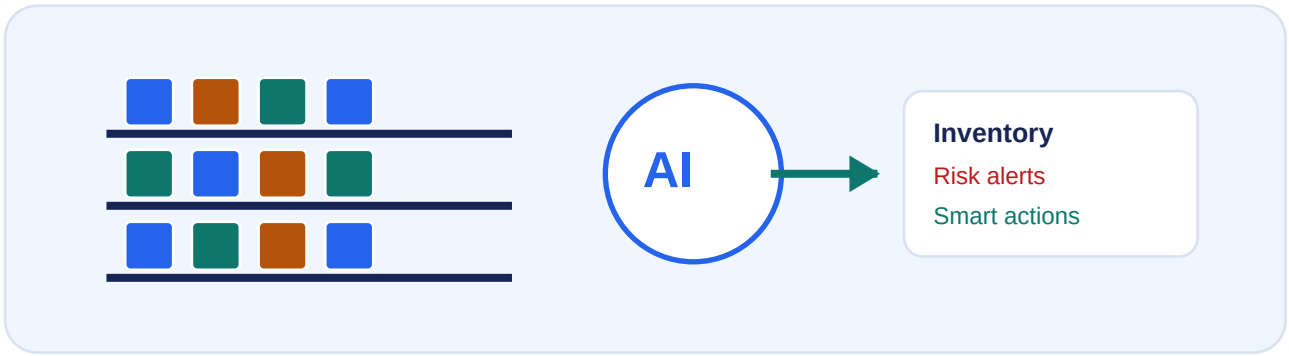
Mercura AI

An AI-Driven Inventory Intelligence System for Merchants



Mercura AI is an intelligent inventory management system that helps shop owners understand what products to buy, what products are not selling, and how to avoid losing money on unsold stock.

Imagine a small supermarket, clothing store, or mobile accessories shop. The owner purchases products to sell to customers. But sometimes, products remain on the shelves for months because customers are no longer interested in them.

The owner's money is now stuck in those products. This is called **dead stock**.

Mercura AI tries to solve this problem by analyzing sales and inventory data, predicting which products might become dead stock, and recommending what the merchant should do next.

1. Understand the problem with a real-life example

Imagine a clothing shop owner who purchases 100 winter jackets at ■1,000 each.

Initial purchase

■1,00,000
100 jackets purchased at ■1,000 each.

After two months

70 jackets
Only 30 sold; 70 remain unsold.

The shop owner has ■70,000 worth of purchase cost tied up in the remaining jackets.

The owner now faces several problems:

- Should they wait for more customers?
- Should they offer a discount to sell the jackets?
- Should they stop ordering more jackets?
- What products should they purchase instead?

Without proper sales analysis, the owner might continue purchasing jackets and end up with even more unsold stock.

This is where Mercura AI helps.

Instead of simply displaying the number of jackets remaining, the system analyzes the situation and might recommend:

- **High risk:** Winter jacket sales are slowing down.
- **Recommended action:** Offer a suitable discount to clear excess stock.
- **Purchase advice:** Avoid reordering until existing stock decreases.
- **Financial insight:** ■70,000 of purchase cost is tied up in unsold jackets.

The exact recommendations would depend on actual sales, margins, inventory age, and seasonal demand data.

2. How Mercura AI works

1	**Step 1: Collect business data** The merchant enters product details, purchase quantities, sales, prices, costs, and stock levels. Data can also be captured through voice input or invoice scanning.
2	**Step 2: Analyze inventory** The system checks how quickly products sell, how long they have remained in stock, their profit margins, and whether sales are increasing or decreasing.
3	**Step 3: Predict future problems** Machine learning models estimate which products may become dead stock, forecast future demand, and identify products that may need replenishment.
4	**Step 4: Recommend actions** Mercura AI explains the problem and suggests actions such as discounting slow-moving products, delaying purchases, or reordering products that are selling well.
5	**Step 5: Display insights on a dashboard** The merchant sees important alerts, product risk levels, expected cash recovery, sales performance, and recommended next steps.

In simple words, Mercura AI follows this process: Collect data → Analyze → Predict → Recommend → Take action.

3. Main features explained with examples

1. Dead Stock Risk Prediction

Identifies products that are selling slowly and may remain unsold.

Example: A shop has 50 phone cases for an outdated model, but only one has sold in the past month. Mercura flags them as high risk.

2. Smart Reorder Suggestions

Recommends when and how much stock to purchase, considering demand and the time suppliers need to deliver.

Example: If a popular product sells 10 units daily and only 20 units remain, Mercura may recommend reordering before the stock runs out.

3. Profit and Demand Analysis

Separates products that sell frequently from products that actually generate good profits.

Example: Product A sells 100 units with ■10 profit per unit, while Product B sells 40 units with ■50 profit per unit. A earns ■1,000, while B earns ■2,000. Product B generates more total profit despite selling fewer units.

4. Festival and Event Demand Alerts

Uses seasonal patterns and relevant events to help merchants prepare for changing demand.

Example: A gift shop may experience higher demand for gift boxes and decorations before Diwali. Mercura can suggest preparing additional stock based on historical sales and available demand information.

5. Cash Flow and Cash Recovery Insights

Shows how much money is tied up in stock and estimates how much cash might be recovered by selling it.

Example: If unsold products cost ■20,000 to purchase, Mercura can estimate potential recovery under different selling prices and discounts. The actual amount recovered depends on what customers pay.

6. Easy Data Entry Using Voice and OCR

Reduces the effort required to maintain records.

Voice: “Sold five packets of biscuits today.” OCR: Scan an invoice to extract product names, quantities, and prices. Quick entry: Manually enter a product or sale. OCR means Optical Character Recognition — technology that reads text from images or scanned documents.

4. What makes Mercura AI different from a normal inventory app?

A traditional inventory application primarily tells the merchant what has happened. Mercura AI aims to explain what is happening, predict what might happen next, and suggest what to do about it.

Question	Traditional inventory app	Mercura AI
How much stock remains?	Shows current quantity	Shows quantity and inventory health
Which products are not selling?	Displays sales history	Identifies slow-moving products and estimates risk
What should I purchase?	May show low-stock alerts	Recommends reorder quantities using demand and supplier lead time
What should I discount?	Merchant decides manually	Suggests candidates based on stock age, demand, and margins
What about future demand?	Mainly historical reports	Forecasts demand when sufficient data is available
What should I do next?	Mostly provides information	Provides recommendations with explanations

The important difference is **decision support**. Mercura AI does not merely show numbers; it attempts to turn those numbers into useful business decisions.

5. Important technical terms you should understand

Sales velocity

How quickly a product sells over a period of time.
Example: If 60 bottles sell in 30 days, sales velocity is 2 bottles per day.

Inventory age

How long a product has remained in stock since it was received or purchased.
Example: A product sitting unsold for 120 days may need attention, depending on its normal selling cycle.

Profit margin

The percentage of the selling price that remains as profit after the relevant costs are accounted for.
Example: If a product sells for ■100 and its cost is ■70, its gross profit is ■30 and its gross margin is 30%.

Supplier lead time

The time between placing an order with a supplier and receiving the products.
Example: If delivery takes five days, the merchant must consider those five days when deciding when to reorder.

Demand forecasting

Estimating how much of a product customers may buy in the future, using historical sales and other relevant factors.

Machine learning

A branch of AI in which a computer learns patterns from data and uses them to make predictions.
Example: A model learns that sales of a certain product decline after a festival and uses that pattern to estimate future demand.

6. How does the AI actually make decisions?

This is an important part of understanding the project. Mercura AI combines business calculations with machine learning.

For example, imagine a product with the following details:

Product analysis example	Value
Current stock	80 units
Average daily sales	2 units
Days of stock remaining at current sales rate	40 days
Recent sales trend	Declining
Inventory age	75 days

Possible AI recommendation
Flag the product for review, consider a targeted discount, and postpone reordering until the excess stock is reduced.

The system reaches this recommendation through several types of analysis:

1. **Business rules and calculations:** Calculate sales velocity, stock coverage, inventory age, and gross profit.
2. **Risk scoring:** Combine relevant factors to estimate how risky the inventory is.
3. **Machine learning predictions:** Learn from historical data to estimate future demand or the likelihood of slow-moving stock.
4. **Recommendation logic:** Use the results to suggest actions, explaining why they may be appropriate.

An important distinction: not every part needs machine learning. Calculating remaining stock or profit is ordinary programming and mathematics. Machine learning becomes useful when the system needs to learn patterns or predict uncertain future outcomes.

Also, a product with 40 days of stock is not automatically dead stock. Its risk depends on factors such as seasonality, expected demand, shelf life, margins, and supplier conditions.

7. Technology stack — what each technology does

React.js + Tailwind CSS — Frontend

Creates the screens that merchants interact with, including the dashboard, inventory tables, charts, and alerts.

Python + FastAPI — Backend

Handles requests, validates business data, runs inventory calculations, communicates with the database, and connects to the AI models.

PostgreSQL — Database

Stores merchants, products, inventory quantities, sales records, purchases, and other business information.

scikit-learn — Machine learning

Provides tools for building and evaluating models for inventory risk classification, demand prediction, and other suitable predictive tasks.

Think of it this way: **React is what the merchant sees, FastAPI is the system's working engine, PostgreSQL is its memory, and machine learning helps it recognize patterns and make predictions.**

These are the proposed technologies in your project description. Their inclusion does not automatically mean every feature or AI model has already been implemented.

8. What would a merchant see on the dashboard?

What is happening?

Overall sales, inventory value, profitable products, stock levels, and cash tied up in inventory.

What is risky?

Products with declining sales, excessive stock, aging inventory, or a high likelihood of becoming dead stock.

What should be done?

Products to discount, items to reorder, purchases to postpone, and inventory that needs immediate attention.

The goal is to make the dashboard easy enough for a small shop owner to understand without needing technical or data-analysis knowledge.

9. Which features should be built first?

Your description includes several core features and some advanced additions. For a practical first version, it makes sense to prioritize them in phases.

Phase	Features
Phase 1: Core system	Merchant login, product and stock entry, sales recording, inventory dashboard, dead stock risk scoring
Phase 2: Intelligence	Smart reorder suggestions, profit analysis, demand forecasting, cash recovery estimates
Phase 3: Convenience	Voice entry, invoice scanning using OCR, festival-based alerts
Phase 4: Automation	Supplier order requests, digital ledger, expanded cash-flow tracking

For the first version, you can demonstrate useful inventory intelligence using sample sales data and transparent risk-scoring rules. You do not need a complicated AI model for every feature.

For machine learning predictions, the quality of the results will depend on the quantity and reliability of the data available. If a new merchant has little historical data, Mercura can start with basic calculations and rules, then improve its predictions as more records accumulate.

10. The entire project in one simple example

Imagine a merchant logs in to Mercura AI in the morning.

1. The dashboard shows that sales of a particular product have fallen significantly.
2. Mercura checks the remaining quantity, inventory age, profit margin, and historical sales.
3. The system flags the product as high risk and explains the reasons.
4. It recommends a discount and advises against purchasing more of that product for now.
5. At the same time, it identifies another product selling quickly and recommends reordering it based on expected demand and supplier lead time.
6. The merchant reviews the recommendations and decides what to do.

The result is a merchant who can make more informed purchasing and pricing decisions, reduce the amount of money stuck in slow-moving stock, and potentially improve profitability.

In one sentence: Mercura AI is like a smart business assistant for shop owners that monitors inventory, warns them about products that may not sell, forecasts demand, and recommends actions to protect their money and improve profits.

Note on visuals: This PDF recreates the response's structure and illustrative imagery using embedded vector illustrations. The original chat images were dynamically rendered and were not available as separate image files to embed byte-for-byte.